Survey-weighted regression with state-level exposures

A methods note on complex-sample and state-clustered inference

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1 Recommendation in brief

For repeated cross-sectional survey data whose **scientific target is the association between a state-level exposure and individual outcomes in the US population represented by the survey**, the recommended **primary analysis** is `survey::svyglm()` with the **actual** sampling weights, primary sampling units (PSUs), and strata. We don't think that it makes sense to replace the original PSU identifier with `statefips` in the survey-design declaration simply because the exposure varies by state.

As a **sensitivity analysis**, fit the *same weighted regression* using `fixest::feols()` with state-clustered standard errors. This probes a different source of uncertainty—within-state dependence across observations, survey PSUs, and years—but does **not** preserve the full sampling-design variance calculation. The two standard errors should not be regarded as interchangeable, nor should one automatically choose the larger (Lumley and Scott, 2017; Cameron and Miller, 2015; Abadie et al., 2023).

1.1 Scientific target and regression specifications

Let i index an individual sampled in state s in survey period t; let A_{st} be an exposure measured at the state-period level, Y_{ist} an individual outcome, and $\mathbf{C}_{ist}$ observed covariates.

The proposed **pooled association model** (without state or time fixed effects) is

$$Y_{ist} = \alpha + \beta_{\text{pool}} A_{st} + \mathbf{C}_{ist}^{\top}\boldsymbol{\gamma} + \varepsilon_{ist}.$$

With $\mathbf{X}_{ist} = (1, A_{st}, \mathbf{C}_{ist}^{\top})^{\top}$, the estimand for a linear regression can be expressed as the exposure coefficient in the **population least-squares projection**:

$$\boldsymbol{\theta}_{\mathcal{P}} = \arg\min_{\boldsymbol{\theta}} \sum_{(i,s,t)\in\mathcal{P}} q_t \big(Y_{ist} - \mathbf{X}_{ist}^{\top}\boldsymbol{\theta}\big)^2.$$

Here q_t describes the intended relative weighting of population periods; it must be reflected in the construction/scaling of pooled survey weights. The pooled coefficient combines information from between-state and within-state/time exposure contrasts.

For comparison, a **two-way fixed-effects specification** is

$$Y_{ist} = \alpha_s + \lambda_t + \beta_{\text{FE}} A_{st} + \mathbf{C}_{ist}^{\top}\boldsymbol{\gamma} + \varepsilon_{ist}.$$

Provided A_{st} changes sufficiently within states after accounting for year effects, β_{FE} is identified without following the same individuals over time. These are different regression specifications and generally estimate different associations. Neither fixed effects nor clustering substitutes for the other.

2 What `svydesign()` and `svyglm()` do

The function `svydesign()` declares the sampling structure (it does **not** by itself fit a model). In particular, `ids` refers to the **sampling PSUs**, `strata` to sampling strata, and `weights` to sampling weights. See the survey documentation.

For a Gaussian identity-link regression (as proposed for the linear probability model in this application), `svyglm()` estimates coefficients by solving the weighted score equations

$$\sum_{(i,s,t)\in\mathcal{S}} w_{ist}\mathbf{X}_{ist}\left(Y_{ist}-\mathbf{X}_{ist}^{\top}\widehat{\boldsymbol{\theta}}\right)=\mathbf{0},$$

which yield

$$\widehat{\boldsymbol{\theta}}=\left(\sum_{\mathcal{S}} w_{ist}\mathbf{X}_{ist}\mathbf{X}_{ist}^{\top}\right)^{-1}\left(\sum_{\mathcal{S}} w_{ist}\mathbf{X}_{ist}Y_{ist}\right).$$

The **design-based** standard errors reflect variability from hypothetical repetitions of the *actual probability-sampling procedure*, treating finite-population outcomes and observed state exposures as fixed. Define $\mathbf{A}=\sum_{\mathcal{S}} w_i\mathbf{X}_i\mathbf{X}_i^{\top}$ and PSU-level weighted residual-score totals $\mathbf{U}_{hj}$, where h indexes the original sampling strata and j indexes sampled PSUs within each stratum. Under the standard first-stage with-replacement Taylor approximation (no finite population correction),

$$\widehat{\mathrm{Var}}_{\mathrm{design}}(\widehat{\boldsymbol{\theta}})=\mathbf{A}^{-1}\underbrace{\left[\sum_h \frac{n_h}{n_h-1}\sum_{j=1}^{n_h}(\mathbf{U}_{hj}-\overline{\mathbf{U}}_h)(\mathbf{U}_{hj}-\overline{\mathbf{U}}_h)^{\top}\right]}_{\mathbf{B}_{\mathrm{design}}}\mathbf{A}^{-1}.$$

Thus, the *meat* of the sandwich measures **between-PSU variation within sampling strata**. Finite population corrections, replicate-weight designs, and single-PSU strata require survey-specific handling. Crucially, `svyglm()` already supplies robust design-based standard errors; see its documentation and Lumley and Scott (2017).

3 What weighted `feols()` with state clustering does

With the **same estimation sample, regression formula, and weights**, `feols()` produces the same weighted least-squares coefficient estimates, up to numerical tolerances. Supplying `vcov = ~statefips` changes **the covariance estimator, not the coefficients**.

Let the weighted residual-score total for state s be

$$\mathbf{U}_s=\sum_{(i,t)\in s} w_{ist}\mathbf{X}_{ist}\left(Y_{ist}-\mathbf{X}_{ist}^{\top}\widehat{\boldsymbol{\theta}}\right).$$

The conventional state-clustered covariance has the general form

$$\widehat{\mathrm{Var}}_{\mathrm{state}}(\widehat{\boldsymbol{\theta}})=\mathbf{A}^{-1}\underbrace{\left[c\sum_{s=1}^{S}\mathbf{U}_s\mathbf{U}_s^{\top}\right]}_{\mathbf{B}_{\mathrm{state}}}\mathbf{A}^{-1},$$

where c is a software-dependent finite-sample correction. This permits arbitrary dependence **within** state, including across PSUs and survey years, while relying on assumptions justifying inference across state clusters. It does **not** use the original survey strata and their sampling-specific covariance structure. See Cameron and Miller (2015), Abadie et al. (2023), and the `fixest` documentation.

The distinction matters especially when all 50 states and DC are observed: states are not themselves a probability sample of US states. A state-clustered analysis can still be useful under an appropriate model- or assignment-based justification; observing a state-level exposure alone does not establish that justification (Abadie et al., 2023).

4 Practical comparison

Feature	`svyglm()` with actual design	Weighted `feols()` clustered by state
Regression coefficients	Population-weighted fit	Same if data, weights and formula agree
Covariance basis	Sampling PSUs **within** sampling strata	Weighted score totals **by state**
Uses original survey strata	Yes	No
Permits arbitrary within-state covariance across different PSUs	Not automatically in a stochastic state-disturbance model	Yes, under state-cluster inference assumptions
Primary use here	**Population descriptive inference under probability sampling**	**Sensitivity analysis** for state-level dependence

Neither covariance estimator automatically captures *both* probability-sampling variability and a separately posited stochastic state-level process. If results differ appreciably, the appropriate response is to clarify the inferential target and investigate a combined framework, not to select whichever standard error is largest.

5 R implementation

The following code is **illustrative**. It assumes a continuous outcome, a common complete-case analysis sample, valid pooled weights, and PSU/stratum identifiers prepared according to the survey's instructions for combining years. For logistic or count outcomes, `svyglm()` remains available with the relevant family, but `feols()` is **not** the same point estimator; use an appropriate corresponding GLM for a valid comparison.

```r
library(survey)
library(fixest)

vars <- c("y", "state_exposure", "x1", "x2", "statefips",
          "year", "psu", "strata", "weight")
analysis_dat <- dat[stats::complete.cases(dat[, vars]), ]

# Check pooled weights and survey documentation before proceeding.
# PSU and strata IDs may need to be made wave-specific.

original_design <- svydesign(
  ids = ~psu,
  strata = ~strata,
  weights = ~weight,
  nest = TRUE,
  data = analysis_dat
)

# Primary pooled analysis: no state or year fixed effects
m_pool_svy <- svyglm(
  y ~ state_exposure + x1 + x2,
  design = original_design,
  family = gaussian()
```

```r
)

# Sensitivity: identical mean model and weights; state-clustered SE
m_pool_state <- feols(
  y ~ state_exposure + x1 + x2,
  data = analysis_dat,
  weights = ~weight,
  vcov = ~statefips
)

# Optional second substantive specification: state and year FE
m_fe_svy <- svyglm(
  y ~ state_exposure + x1 + x2 + factor(statefips) + factor(year),
  design = original_design,
  family = gaussian()
)

m_fe_state <- feols(
  y ~ state_exposure + x1 + x2 | statefips + year,
  data = analysis_dat,
  weights = ~weight,
  vcov = ~statefips
)

# Check equality of point estimates for the exposure
c(survey = coef(m_pool_svy)["state_exposure"],
  state_clustered = coef(m_pool_state)["state_exposure"])

# Compare standard errors
c(survey = sqrt(vcov(m_pool_svy)["state_exposure", "state_exposure"]),
  state_clustered = sqrt(vcov(m_pool_state)["state_exposure", "state_exposure"]))

# Survey design degrees of freedom
survey::degf(original_design)
```

5.1 Checks before interpretation

1. **Weight harmonization:** Confirm that pooled survey weights reproduce the intended population-period weighting, and that any official multi-year weight instructions have been followed.
2. **Design identifiers:** Verify whether PSU and stratum IDs repeat across years; distinguish reused codes from genuinely repeated sampling units.
3. **Survey design details:** Use supplied replicate weights or FPC information if required; inspect certainty PSUs and single-PSU strata.
4. **Sample alignment:** Fit all comparisons on the same complete cases and confirm numerical equivalence of the pooled regression coefficients.
5. **Interpretation:** Pooled and fixed-effects exposure coefficients can differ because they exploit different contrasts; differences are not simply due to variance estimation.

6 Recommended reporting language

> We fitted survey-weighted regression models to estimate associations between state-level exposure and individual-level outcomes in the US population represented by the repeated cross-sectional survey. Our primary inference used Taylor-linearized standard errors from the original multistage survey sampling design, incorporating sampling weights, primary sampling units, and sampling strata. As a sensitivity analysis, we fitted the same weighted regression specifications with standard errors clustered at the

state level, allowing dependence among observations from the same state across survey units and years. This sensitivity analysis addresses a distinct inferential framework and does not replace the original survey-design variance estimator.

7 References

Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics, 138*(1), 1–35. https://doi.org/10.1093/qje/qjac038

Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics, 119*(1), 249–275. https://doi.org/10.1162/003355304772839588

Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources, 50*(2), 317–372. https://doi.org/10.3368/jhr.50.2.317

Lumley, T., & Scott, A. (2017). Fitting regression models to survey data. *Statistical Science, 32*(2), 265–278. https://doi.org/10.1214/16-STS605

Software documentation: `survey::svyglm()` https://r-survey.r-forge.r-project.org/pkgdown/docs/reference/svyglm.html; `survey::svydesign()` https://r-survey.r-forge.r-project.org/pkgdown/docs/reference/svydesign.html; `fixest::feols()` https://lrberge.github.io/fixest/reference/feols.html.